7 **(a)** Radioactive decay is both *random* and *spontaneous*. State what is meant by:

(i) random

.....

.....[1]

(ii) spontaneous

.....

.....[1]

(b) Polonium-193 decays into lead-189 and an alpha particle. The binding energy per nucleon of the particles are given in Table 7.1.

particle	binding energy per nucleon / MeV
$^{193}_{84}Po$	7.774
$^{189}_{82}Pb$	7.826
$^{4}_{2}He$	7.074

Table 7.1

(i) Determine the energy, in MeV, released when a nucleus of polonium-193 decays into a nucleus of lead-189.

energy = ... MeV [2]

(ii) At an instance in time, a block of polonium was found to be emitting energy at a rate of 16.1 W. Calculate the activity of the polonium block.

activity = ... Bq [2]

(iii) The half-life of polonium-193 is approximately 420 ms. Calculate the time it will take a pure sample of polonium to decay to 1.0% of its initial amount.

time = ... s [2]

[Total: 8]